

BINUS University

Undergraduate Program <i>Regular/Smart Program/Global Class*)</i>		Term : Odd/Even/Short *)
☐ Mid Exam ☐ Final Exam ☐ Short Term Exam ☒ Others Exam : Mid Exam PPT1 7		Academic Year : 2019 / 2020
Faculty / Dept. : School of Computer Science		Student ID :
Code / Course : COMP6047 – Algorithm And Programming		
Class : LAT8 (111AT)		N a m e :
Shift : Morning		
Day/ Date : <i>Wednesday</i> , August 28 th 2019		Signature :
Time : 08:30 – 10:10 (..... Minutes)		
Lecturer : D5797 – Felix Jingga, S.Kom.		
Exam Type : Open Book/Close Book/Open e-Book/ Submit Project/Oral Test *)		
Equipment : Exam Booklet/Calculator/Dictionary/ Laptop/Tablet/Smartphone/ Drawing Paper – A3/Drawing Paper – A2 *)		
*) <i>Strikethrough the unnecessary items</i>		
<i>Please insert the test paper into the exam booklet and submit both papers after the test!!!</i> <i>The penalty for CHEATING is DROP OUT!!!</i>		

Rules:

Students **ARE NOT ALLOWED** to bring:

- Any digital materials (e.g., e-books, softcopy codes, etc.)
- Any digital devices (e.g., flash disks, external hdd, **apple watch or any other smart watches**, etc.)
- Any other electronic devices (e.g., smartphone, handphone, calculator, laptop, etc.)
- Any communication devices should be turned off during this exam.

Fair Examination

A student may be zero-scored for any cheat or attempt of cheat (submitting a solution which is not made by yourself is considered as a very serious cheating) and will be processed through University's policy regarding cheating. A student may also be excused from the exam room for any activity that jeopardized the exam (e.g., **hacking the exam system**, dislodging extension cords, distracting behavior, etc.). All students should uphold the honesty and the spirit of fair examination.

Problem A (25%)
Print a Sequence

Bibi has a number N and he'd like to write a sequence of 10 integers that are multiples of N . As a very good friend of Bibi, help him by writing a code to do that.

Format Input

The first line of the input consists of a single integer T , denoting the number of test cases. After that, T test case followed. Each test case consists of an integer N .

Format Output

For each test case, output a single line containing a sequence of 10 integers that are multiples of N . Each integer is separated by a space.

Constraints

$$1 \leq T \leq 100$$

$$1 \leq N \leq 100$$

Sample Input	Sample Output
3	7 14 21 28 35 42 49 56 63 70
7	4 8 12 16 20 24 28 32 36 40
4	13 26 39 52 65 78 91 104 117 130
13	

Notes

Even though it's not stated explicitly, you should know by now that excessive space / newline is treated as **WRONG ANSWER**.

*The quality of a person's life is in direct proportion to their commitment
to excellence, regardless of their chosen field of endeavor.*
– Vince Lombardi

Problem B (25%)

Run-Length Encoding

Lili learns a new technique named “Run-Length” Encoding. This technique was widely used as a compression technique in the past. “Run-Length” Encoding can be described by series of following example:

- “WWWBWWWWWW” (without quotes) will become “3W1B6W” (without quotes)
- “Wweee aaarreeeee fffrriieeenndsss” will become “1W1w3e 3a2r5e 3f2r2i3e2n1d3s”

You’ll be given a sentence S, and you need to output the sentence S in the “Run-Length” Encoding format.

Format Input

The first line of the input consists of a single integer T, denoting the number of test cases. After that, T lines followed. Each of that T lines denotes a single test case that contains a string S.

Format Output

For each test case, you should output the case number starting from 1, and in the next line the string S in “Run-Length” Encoding format. Please look at the sample output for reference.

Constraints

$$1 \leq T \leq 10$$

$$1 \leq |S| \leq 100.000$$

|S| is the length of the string

S will only contain alphabets (a-z / A-Z) and each word in a sentence will be separated by a single space.

Sample Input	Sample Output
3 WWWBWWWWWW Heelllloooooo biiiiibiiiiiiiiiiii Wweee aaarreeeee fffrriieeenndsss	Case #1: 3W1B6W Case #2: 1H3e3l6o 1b6i1b12i Case #3: 1W1w3e 3a2r5e 3f2r2i3e2n1d3s

Notes

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No matter how good you get you can always get better, and that's the exciting part.
– Tiger Woods

Problem C (25%)

A Prime Number

This is an easy classical problem for you. Given an integer N , you'll need to output whether that integer N is a Prime Number or not. A number N is called prime if that number's divisors are exactly 2, the number 1 and itself (N).

Format Input

The first line of the input consists of a single integer T , denoting the number of test cases. After that, T test case followed. Each test case consists of an integer N .

Format Output

For each test case, you should output the case number starting from 1, followed by a "YES" or "NO". "YES" is when the number N is a Prime Number, and "NO" otherwise. Please look at the sample output for reference.

Constraints

$$1 \leq T \leq 1000$$

$$1 \leq N \leq 4294967295$$

Sample Input	Sample Output
10	Case #1: NO
1	Case #2: YES
3	Case #3: NO
4	Case #4: YES
5	Case #5: YES
11	Case #6: NO
6	Case #7: YES
7	Case #8: NO
28	Case #9: YES
29	Case #10: NO
49	

Notes

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You **won't get 100** for this problem **unless** your solution is **really efficient**.

For the things we have to learn before we can do them, we learn by doing them.
– Aristotle

Problem D (25%)

Reverse Number

Jojo wants you to make him a program that can output a number in its reverse polynomial form. For example, the number 123 can be expressed as “ $3 \times 1 + 2 \times 10 + 1 \times 100$ ”, and 6712 can be expressed as “ $2 \times 1 + 1 \times 10 + 7 \times 100 + 6 \times 1000$ ”.

Format Input

The first line of the input consists of a single integer T , denoting the number of test cases. After that, T test case followed. Each test case consists of an integer N .

Format Output

For each test case, you should output the case number starting from 1, and in the next line is the number in its Reverse Polynomial Form. Please look at the sample output for reference.

Constraints

$$1 \leq T \leq 1000$$

$$1 \leq N \leq 1.000.000.000$$

Sample Input	Sample Output
3 123 6712 1002	Case #1: $3 \times 1 + 2 \times 10 + 1 \times 100$ Case #2: $2 \times 1 + 1 \times 10 + 7 \times 100 + 6 \times 1000$ Case #3: $2 \times 1 + 0 \times 10 + 0 \times 100 + 1 \times 1000$

Notes

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Strength does not come from winning. Your struggles develop your strengths. When you go through hardships and decide not to surrender, that is strength.
– Arnold Schwarzenegger

-- Selamat mengerjakan --